Integration of 3D geological model and architectural model for design and construction of bored tunnel in Singapore

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ABSTRACT

This study focuses on the development of a three-dimensional (3D) geological model based on the exploratory borehole data and its integration with architectural model of the Riviera Interchange Station and Tunnels for the Cross Island Line Punggol Extension of Mass Rapid Transit (MRT) in Singapore. The integrated model was used during the design and construction planning stages to support decisions related to excavation volume estimation and tunnelling strategy. One key achievement is the ability to calculate the anticipated excavation volume by geology for each 1.5 to 2-metre ring of tunnel lining, which assisted in aligning excavation methods with ground conditions. However, challenges remain, particularly in addressing uncertainties inherent to 3D geological modelling. Due to the linear nature of tunnels, borehole investigations are typically aligned longitudinally, resulting in limited cross-sectional data and more uncertainty in geological predictions perpendicular to the tunnel alignment. Future refinements may help improve the reliability of geological modelling and its application in tunnel construction planning.

Keywords: 3D geological model, 3D architectural model, underground railway, bored tunnel, tunnel boring machine (TBM), level of development (LOD)

1 INTRODUCTION

The construction of underground infrastructure in urban areas faces numerous challenges, such as limited space, complex ground conditions, and the need to minimise disruption to the existing environment. To address these challenges, the integration of three-dimensional (3D) geological models with architectural models has become increasingly important. In this study, we present the implementation of such an approach for the design and construction of a bored tunnel as part of the Cross Island Line Punggol Extension of Mass Rapid Transit (MRT) in Singapore. The objective is to demonstrate how digital modelling technologies can enhance planning and improve safety.

2 PROJECT DESCRIPTIONS

The Riviera Interchange Station and associated tunnel works are part of the Cross Island Line - Punggol Extension (P103), a new underground MRT line currently under development in Singapore. This project involves the construction of a new underground interchange station adjacent to the existing above-ground Riviera Station of the Light Rail Transit (LRT). The station box will be constructed at an approximate excavation depth of 35 metres below ground level. In addition, approximately 4.3 km of bored tunnels will be constructed. The bored tunnels will be excavated using Earth Pressure Balance (EPB) Tunnel Boring Machine (TBM), with an outer diameter of 12.66 metres, described by Yugami (2023). The tunnels will run at overburden depths ranging from 7 to 34 metres, depending on the alignment and ground conditions.

A cripple siding will be constructed at approximately 0.9 km from the eastern boundary of the P103 construction area. The siding will have a total length of approximately 200 metres, with a depth of 25 metres. At both ends of the cripple siding, two launch shafts will be constructed to facilitate the lowering of the TBM. Each shaft will have a length of 87 metres, a width of 24 metres, and a depth of 28 metres.

The site is situated in an urban area including a residential district of 40,000 households, public facilities such as schools and a shopping mall, as well as critical infrastructure like a fire department, elevated railways, and military-related structures. Given the highly urbanised surroundings and the complex subsoil profile, careful engineering design and construction are essential to minimise risks to nearby buildings, roads, and existing utility services.

3 GEOLOGICAL AND GEOTECHNICAL CONDITIONS

The project alignment is situated within the old river

valleys of Singapore, characterised by marginal-marine, coastal, and fluvial sediments dating from the late Pleistocene to Holocene age, collectively known as the Kallang Group proposed by Leslie et al. (2021). The Kallang Group represents the youngest stratigraphic unit in Singapore and unconformably overlies the irregular upper surface of the Bedok Formation. The Bedok Formation, predating the Pleistocene, comprises weakly consolidated to unconsolidated beds of gravel, sand, silt and clay. The beds are generally heavily altered, oxidised and cemented to varying degrees.

The subsoil conditions along project alignment consists of very soft to soft marine (M) and estuarine clay (E) and firm to stiff alluvial clay (F2) of the Kallang Group, with thicknesses varying from 5 to 15 metres. The Kallang Group also comprises loose to medium dense alluvial sands and silty sands (F1). Further down, the ground transitions into the Bedok Formation, which consists of stiff to very stiff sandy silt and medium dense to very dense silty sand, classified as O(E) to O(A) based on Standard Penetration Test (SPT) N-value. The general subsoil profile along the tunnel alignment, as generated by the 3D geological model, is presented in Fig. 1.

The tunnelling depth occasionally coincides with the alluvial sand layer (F1) of the Kallang Group, which is highly permeable and poses a potential risk of groundwater inflow and instability during construction. Additionally, at the interface between the Kallang Group and the Bedok Formation, the irregular surface of the Bedok Formation results in sudden variations in soil strength, incurring the risk of excavation difficulties, TBM performance disruptions, and tunnel instability.

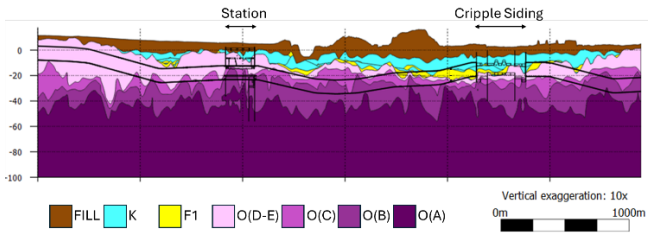


Fig. 1. General subsoil profile along the tunnel alignment.

4 METHODOLOGY

The 3D geological model was generated using Leapfrog Works software by Seequent, which enabled rapid interpolation between borehole logs to produce a continuous and detailed subsurface representation.

The geological modelling workflow is composed of three primary stages: data import, surface generation and 3D model creation. Upon importing data into Leapfrog Works, the contact points for each geological unit are extracted. The surfaces of geological contact are interpolated from the points using an interpolation algorithm called FastRBF (Radial Basis Function), which accelerates the modelling process and enables dynamic updates as new data is integrated. In constructing geological contact surfaces within the 3D geological model, geological interpretations considered whether the interface represents an erosional surface or a depositional contact. Finally, these interpreted surfaces are used to generate the 3D geological model.

The model was developed using data from approximately 400 exploratory boreholes, each ranging from 40 to 50 metres in depth. Boreholes are spaced at intervals of 30 to 50 metres along the overall tunnel alignment and approximately 10 metres within station areas. Out of these, 100 verification boreholes were conducted following the initial soil investigation phase. SPTs were performed at 3-metre intervals, supplemented by various laboratory and in-situ tests. Additionally, terrain data with spacings of 5 to 20 metres, river bathymetric data with 2-metre spacings, and the published geological map (Leslie et al., 2021) were incorporated.

Given the potential risks associated with certain geological formations, the borehole data was reinterpreted and converted into geotechnical layers to mitigate these risks effectively. The modelling units used in the 3D geological model is shown in Table 1.

Table 1. Modelling units in 3D geological model.

Model-ling Units	Singapore Geology Code	Stratigraphy and Lithology
FILL	FILL	Silty/sandy artificial ground
K	M, E, F2	Very soft to firm Kallang Group clay
F1	F1	Loose to medium dense Kallang Group sand
O(D-E)	O(D), O(E)	Silty/sandy Bedok Formation with SPT N less than 30
O(C)	O(C)	Same as above with SPT-N 30 to 50
O(B)	O(B)	Same as above with SPT-N 50 to 100
O(A)	O(A)	Same as above with SPT-N > 100

Subsequently, the architectural model created by the design and construction contractor was imported into the 3D geological model environment. Each tunnel ring, composed of tunnel segments spaced 1.5 to 2 metres, was imported for tunnel alignment. The tunnel segment models feature dimple holes; however, support patterns such as rock bolts and reinforcement are not included. The architectural models of station and cripple siding allow for the measurement of elements, including quantity, size, shape, location, and orientation. The overall and close-up views of 3D integrated model are shown in Fig. 2 and Fig. 3 respectively.

5 APPLICATION TO TUNNEL DESIGN AND CONSTRUCTION

The integrated model provided significant advantages in the planning of tunnel excavation. By calculating the expected excavation volume for each geological unit along 1.5 to 2.0 metre tunnel segments,

it became possible to optimise TBM tunnelling operations. Fig. 4 shows an example of cross-sectional diagram of estimated excavation areas by each geological unit at a tunnel segment, generated from the 3D integrated model. The estimated excavation volumes by each geological unit are tabulated in Table 2 for the overall tunnel alignment.

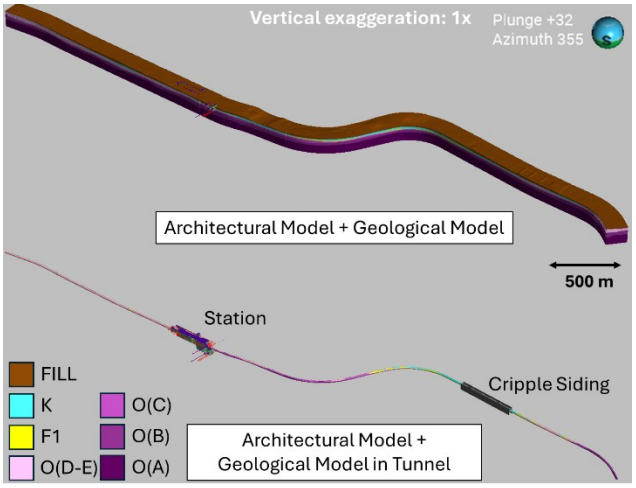


Fig. 2. Overall view of the 3D integrated model.

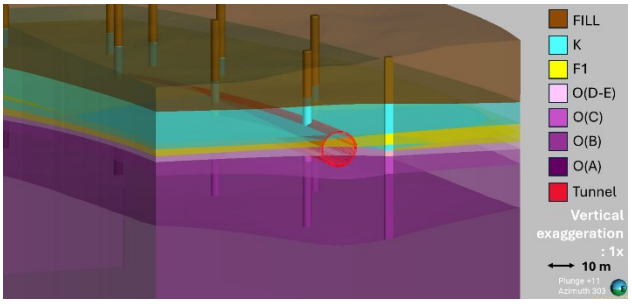


Fig. 3. Close-up of the 3D integrated model.

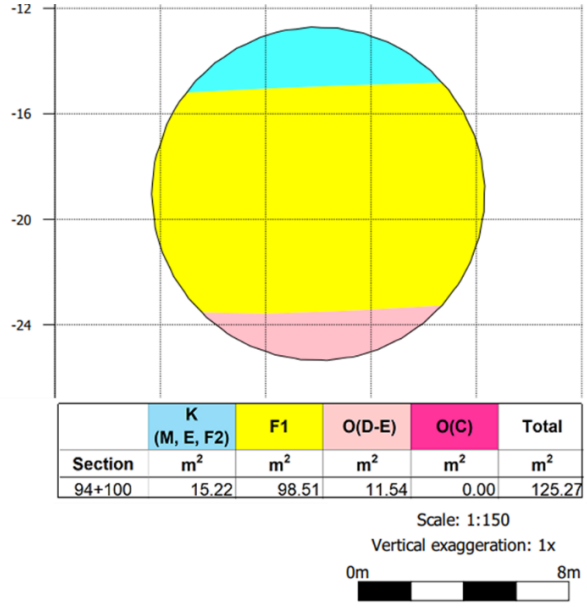


Fig. 4. Cross-sectional diagram of excavation areas at tunnel segment.

Table 2. Estimated excavation volumes for the overall tunnel alignment

Modelling Units	Estimated Excavation Volume (m^3)
K	36,630
F1	16,340
O(D-E)	239,440
O(C)	125,120
O(B)	126,680
O(A)	390

The model also allowed visual comparison between underground structures and subsurface conditions. This helped engineers and geologists identify areas where ground conditions might affect construction and make informed adjustments during design and planning. Fig. 5 depicts the spatial relationship between the underground structures and the F1 layer, which is the alluvial sand layer and poses a potential risk of groundwater inflow and instability during construction. From this figure, the bored tunnels are expected to coincide with the F1 layer in the vicinity of the cripple siding.

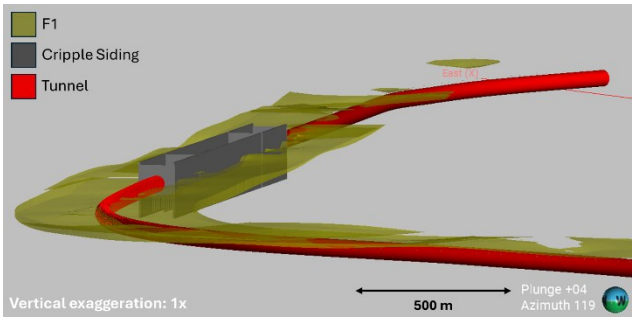


Fig. 5. Spatial relationship between underground structures and the F1 layer.

While the integrated model was useful, its accuracy is limited by uncertainties in geological interpretation, particularly where borehole data are sparse in directions perpendicular to the tunnel alignment. To manage this, it is important to record the sources of data, the assumptions made, and the level of confidence in each part of the model. Sharing this information with designers, contractors, and owners helps maintain a consistent understanding of the risks involved and supports practical decision-making during design and construction.

The excavation volumes could be validated during construction stage, where the excavated volume of each geological units will be recorded during the tunnelling process for each segment.

In integrating geological and architectural models, another difficulty arises from the difference in the Level of Development (LOD) for each model type. Geological models are generally constructed based on limited investigation data to capture large-scale subsurface features, and therefore possess a relatively low level in terms of resolution and geometry. In contrast, architectural models are developed with fine spatial resolution to represent structural and design components

with high precision. As a result, aligning these two models within the same environment can be technically challenging. The development of effective approaches to address this gap remains a subject for future consideration.

6 FUTURE POTENTIAL

The use of an integrated geological and architectural model could contribute to better coordination between design and construction teams. One possible improvement is the refinement of tunnel segment designs based on geological conditions predicted by the model. For example, segment types could be adapted in advance for areas with soft soils or stiffer soils or bedrock, which may improve efficiency and reduce construction risk. Further development of such applications requires improved geological prediction accuracy and closer collaboration among stakeholders involved in tunnelling projects.

7 CONCLUSIONS

The integration of 3D geological and architectural models provided a practical tool for coordinating design and construction planning for the bored tunnels of this MRT project. In particular, the ability to estimate excavation volumes by geological unit helped align tunnelling operations with expected ground conditions. While the model improved visibility into subsurface conditions, limitations remain due to the inherent uncertainties of geological prediction, especially where borehole data are sparse. Continued refinement of modelling techniques, including more comprehensive data collection and clearer representation of uncertainty, will be necessary to improve the reliability and usefulness of such models in future tunnelling projects.

Differences in the LOD between the geological model and the architectural model remains still technical challenge. This case study highlights the differences in LOD observed within an ongoing MRT project in Singapore. The LOD in the 3D geological model should be standardised given the complexity of geology, quantity and quality of soil investigation, proposed structure, project stage, and purpose of modelling, to address and overcome the current technical challenges arising from the integration of 3D geological model and architectural model for design and construction of future underground projects in Singapore.

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